Artificial Intelligence Homework 5

June 1, 2026

Q1: Consider N data points (x_i, y_i), where the y_i's are generated from the x_i's according to the linear Gaussian model in Equation (20.5) of the book. Find the values of θ_1, θ_2, and σ that maximize the conditional log likelihood of the data.

Q2: Consider a single Boolean random variable Y (the "classification"). Let the prior probability $P(Y = \textbf{true}) = \pi$. Let's try to find π, given a training set $D = (y_1, \ldots, y_N)$ with N independent samples of Y. Furthermore, suppose p of the N are positive and n of the N are negative.

1. Write down an expression for the likelihood of D (i.e., the probability of seeing this particular sequence of examples, given a fixed value of π) in terms of π, p, and n.

2. By differentiating the log likelihood L, find the value of π that maximizes the likelihood.

3. Now suppose we add in k Boolean random variables $X_1, X_2, \ldots, X_k$ (the "attributes") that describe each sample, and suppose we assume that the attributes are conditionally independent of each other given the goal Y. Draw the Bayes net corresponding to this assumption.

4. Write down the likelihood for the data including the attributes, using the following additional notation:

   (a) α_i is $P(X_i = \text{true}|Y = \text{true})$.

   (b) β_i is $P(X_i = \text{true}|Y = \text{false})$.

   (c) p_i^+ is the count of samples for which $X_i = \text{true}$ and $Y = \text{true}$.

   (d) n_i^+ is the count of samples for which $X_i = \text{false}$ and $Y = \text{true}$.

   (e) p_i^- is the count of samples for which $X_i = \text{true}$ and $Y = \text{false}$.

   (f) n_i^- is the count of samples for which $X_i = \text{false}$ and $Y = \text{false}$. Hint: consider first the probability of seeing a single example with specified values for $X_1, X_2, \ldots, X_k$ and Y.

5. By differentiating the log likelihood L, find the values of α_i and β_i (in terms of the various counts) that maximize the likelihood and say in words what these values represent.

Q3: Consider the following data set comprised of three binary input attributes (A_1, A_2, and A_3) and one binary output: Use the algorithm in Figure 18.5 (page 702) to learn a decision tree for these data. Show the computations made to determine the attribute to split at each node.

Example	A_1	A_2	A_3	**Output** y
$\mathbf{x_1}$	1	0	0	0
$\mathbf{x_2}$	1	0	1	0
$\mathbf{x_3}$	0	1	0	0
$\mathbf{x_4}$	1	1	1	1
$\mathbf{x_5}$	1	1	0	1

Q4: Suppose you had a neural network with linear activation functions. That is, for each unit the output is some constant c times the weighted sum of the inputs.

1. Assume that the network has one hidden layer. For a given assignment to the weights $\mathbf{w}$, write down equations for the value of the units in the output layer as a function of $\mathbf{w}$ and the input layer $\mathbf{x}$, without any explicit mention of the output of the hidden layer. Show that there is a network with no hidden units that computes the same function.

2. Repeat the calculation in part 1, but this time do it for a network with any number of hidden layers.

3. Suppose a network with one hidden layer and linear activation functions has n input and output nodes and h hidden nodes. What effect does the transformation in part 1 to a network with no hidden layers have on the total number of weights? Discuss in particular the case $h \ll n$.